Anvil-E2 Phase 2: Official-Weight Multimodal RTL Evaluation

Technical Design and Verification Report

16 October 2026 (UTC+8)

Abstract

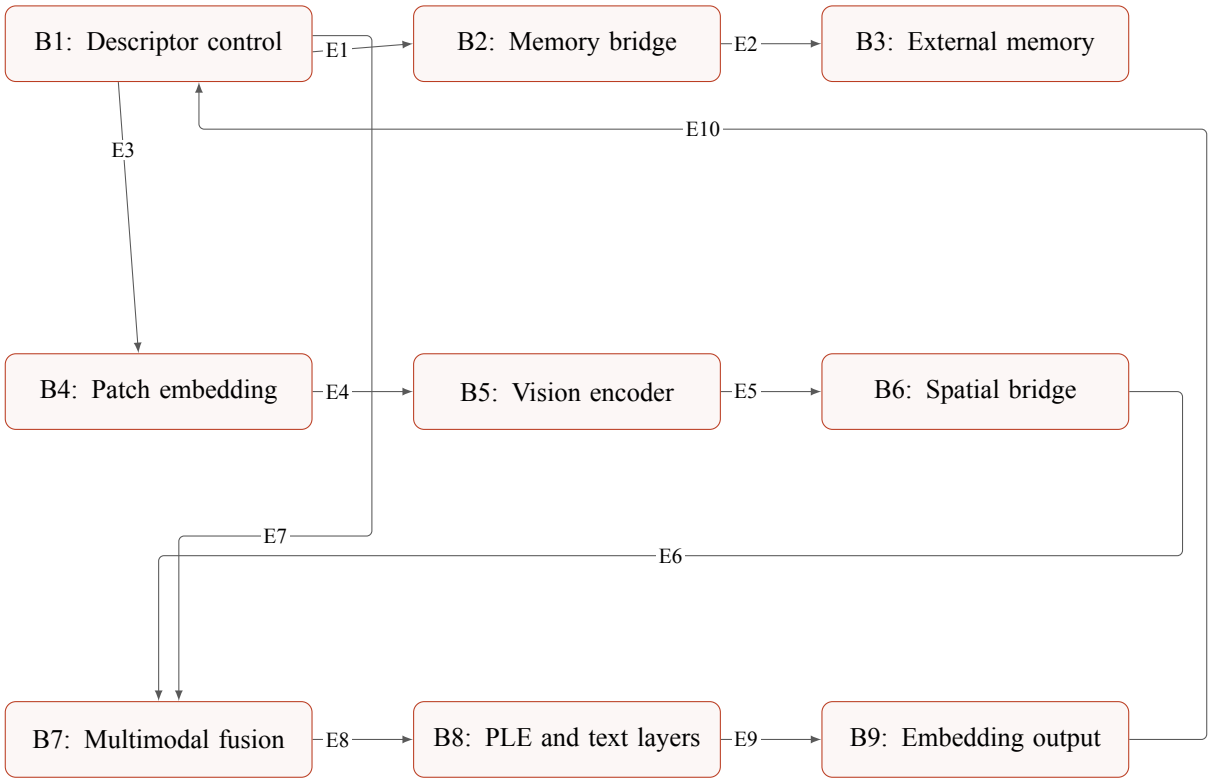
Anvil-E2 is a parameterized SystemVerilog embedding accelerator supporting text-only, vision-only and joint vision-plus-text execution through a shared external-memory interface. Phase 2 integrates directly decoded official BF16 checkpoint slices into independent floating-point software and signed Q8.24 hardware evaluation paths.

All 29 predetermined official-weight cases passed public-interface RTL simulation with zero output-word mismatches against the integer oracle and exact scratch-write histories. Across these cases, minimum cosine similarity was 0.9999999950454138 and maximum absolute component error was 0.000047340989112854, exceeding the required 0.999 similarity and meeting the 0.002 error limit. Functional regression, mutation sensitivity and generic synthesis also passed.

These measurements establish agreement for explicitly remapped, scaled checkpoint subnetworks. They do not establish native HuggingFace full-model equivalence, pretrained application accuracy or physical implementation performance.

1 Architecture and design rationale

The design trades parallel throughput for reusable arithmetic and explicit external-memory materialization. A descriptor controller sequences seven connected processing stages, reusing the vision and text engines across layer iterations. A single-outstanding bridge serializes reads and acknowledged writes. Intermediate tensors reside in a documented scratch arena rather than a model-sized internal frame cache. The diagram separates control and memory service from logical tensor dependencies: stage-to-stage arrows represent scratch-mediated dataflow, not direct streaming wires. Text-only execution enters fusion without the vision stages; vision-only execution still uses framing and the language-side embedding pipeline. Registered clock ownership isolates inactive stages and arithmetic helpers without altering public handshakes. The implementation and memory contract are defined by E001 and E019.



Functional block map, not a gate-level netlist. Edge identifiers match the directed relationships below. Crossings do not denote junctions; separate attachment points preserve arrow directions.

Block	Responsibility
B1: Descriptor control	anvil_e2_top captures and validates one job, registers stage ownership, iterates configured layers and holds completion status.
B2: Memory bridge	anvil_e2_memory serves the selected owner with one outstanding aligned transact ion and a separately retired result.
B3: External memory	Host-provisioned language and vision weights, CHW image data, token metadata and scratch tensors; no unrounded private tensor sidecar.
B4: Patch embedding	anvil_e2_vision_patch_embed projects RGB patches and adds the configured factorized position coordinates.
B5: Vision encoder	anvil_e2_vision_encoder_layer performs reusable RMS, attention, axial rotary and gated feed-forward computation.
B6: Spatial bridge	anvil_e2_vision_pool_proj averages actual pooling participants and projects vision channels into language-width soft tokens.
B7: Multimodal fusion	anvil_e2_multimodal_fusion materializes text-only inputs or BOI, computed IMAGE vectors, EOI and appended prompt metadata.
B8: PLE and text layers	anvil_e2_text_frontend_ple caches initial-vector per-layer inputs; anvil_e2_ hybrid_layer executes scheduled grouped attention, MLP and PLE injection.
B9: Embedding output	anvil_e2_pool_norm applies final language normalization and projection, masked mean, prefix selection and epsilon-free L2 normalization.

Directed relationship	Meaning
E1: B1 to B2	Registered memory-owner selection
E2: B2 to B3	Read/write requests and acknowledged responses
E3: B1 to B4	Image-mode stage start
E4: B4 to B5	Materialized patch vectors
E5: B5 to B6	Final vision-layer vectors
E6: B6 to B7	Projected soft tokens

Directed relationship	Meaning
E7: B1 to B7	Mode and text descriptor
E8: B7 to B8	Fused vectors, IDs, positions and masks
E9: B8 to B9	Final language vectors
E10: B9 to B1	Held stage completion or error

2 Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Clock and asynchronous active-low reset. Asserted reset forces every public output to zero and cancels active work; the environment must discard canceled responses.
job_valid, job_ready, busy	Input / output / output	1 bit each	Capture exactly one descriptor on job_valid && job_ready. Idle after reset release has job_ready=1 and busy=0. No descriptor is accepted while executing or holding completion.
job_mode, job_tokens, job_patch_rows, job_patch_cols, job_output_dim	Input	2, 32, 32, 32, 16 bits	Modes 0/1/2 select text/vision/joint; mode 3 is invalid. Vision-only requires zero supplied text; joint requires positive text count. Output dimension is EMBED_DIM, 2*EMBED_DIM/3, EMBED_DIM/3 or EMBED_DIM/6.
weights_base, vision_weights_base, image_base, tokens_base, positions_base, mask_base, scratch_base	Input	64 bits each	Byte-addressed region bases captured with the descriptor. Active regions must be aligned and nonoverlapping; the host provisions all consumed data and sufficient scratch extent.
mem_req_valid, mem_req_write, mem_req_addr, mem_req_wdata; mem_req_ready	Output; input	1, 1, 64, 32; 1 bit	Aligned little-endian 32-bit memory requests. Request payload remains stable under backpressure. At most one accepted transaction is outstanding.
mem_rsp_valid, mem_rsp_rdata, mem_rsp_error; mem_rsp_ready	Input; output	1, 32, 1; 1 bit	Exactly one response is required for every accepted read or write. A response cannot be accepted on its request-acceptance edge.
emb_valid, emb_data, emb_index, emb_last; emb_ready	Output; input	1, signed 32, 16, 1; 1 bit	Ordered signed Q8.24 embedding words with zero-based indices. Data, index and last remain stable while stalled; a one-element prefix has first index and last on the same beat.
done_valid, done_error; done_ready	Output; input	1, 4; 1 bit	Held completion handshake. Error codes are 0 success, 1 descriptor, 2 fetched metadata, 3 memory response, 4 all-masked pooling, 5 address/extent overflow and 6 arithmetic/internal protocol.

Parameter	Default	Meaning / supported values
VISION_LAYERS, VISION_HIDDEN, VISION_FFN	16, 768, 3072	Positive values, each at most 65535; layer count controls actual engine reuse.
VISION_HEADS, VISION_HEAD_DIM, PATCH_SIZE	12, 64, 16	Positive; VISION_HIDDEN=VISION_HEADS*VISION_HEAD_DIM; head dimension divisible by four; 3*PATCH_SIZE*PATCH_SIZE<=65535.
POOL_KERNEL, VISION_SOFT_TOKENS	3, 280	Positive; descriptor geometry must produce the configured soft-token count, including partial edge windows.

Parameter	Default	Meaning / supported values
TEXT_LAYERS, TEXT_HIDDEN, PLE_DIM, TEXT_FFN	24, 512, 512, 2048	Positive and at most 65535; singleton hidden, PLE and FFN dimensions are supported.
EMBED_DIM	768	6..65535, divisible by six; controls projection rows and the four supported output prefixes.
VOCAB_SIZE, POSITION_EMBEDDING_SIZE	262144, 10240	Positive; determine embedding and position address geometry and metadata validation.
TEXT_Q_HEADS, LOCAL_KV_HEADS, FULL_KV_HEADS	4, 2, 1	Positive; each KV-head count divides the query-head count.
LOCAL_HEAD_DIM, FULL_HEAD_DIM	256, 512	Positive and even; each query-head count times head dimension must fit 65535.
SLIDING_WINDOW, MAX_SEQ_LEN	512, 262144	Positive; bound local eligibility and accepted sequence/position values respectively.
RMS_EPSILON_Q80	Unsigned 128-bit (128'd17 << 56), equal to 12249790 98644774912	Official evaluation selects 128'd1208925819614629175 before execution, the nearest-even conversion of 2^80/1000000. The identical value propagates through every RMS wrapper; L2 ignores it.

3 Behavior and corner cases

3.1 Multimodal computation

Vision executes patch projection, position addition, RMS-normalized Q/K/V, per-head normalization, axial RoPE, bidirectional scaled softmax attention, output residual and gated MLP. Actual-participant spatial means feed the bridge. Fusion scales frame/text embeddings by sqrt(TEXT_HIDDEN), leaves image vectors unscaled and shifts appended positions. Text layers use full attention at zero-based layers 5/11/17/23 and local attention otherwise, with abs(position difference)<SLIDING_WINDOW. RoPE uses the documented vision/local/full theta values 100/10000/1000000.

3.2 Fixed-point arithmetic

External numerical words have 24 fractional bits; signed 48-bit local activations have 40. Let W=2^24, G=2^40, RN denote signed nearest-even division and C clamp to [(-2147483648)<<16,2147483647<<16]. Promotion is q<<16; materialization is signed32-saturating RN(x,2^16). Dense projections use exact 128-bit reductions followed by C(RN(sum,W)) for Q40-by-Q24 products. Residuals use C(a+b), and guarded products use C(RN(a*b,G)).

3.3 Normalization and output

RMS computes D=RN(sum(a*a),count)+RMS_EPSILON_Q80, R=floor(sqrt(D<<64)), t=RN(a<<72,R), and C(RN(t*(W+w),W)); normalization weights therefore multiply by 1+weight. L2 instead uses D=sum(a*a), without epsilon, and returns exact zero for zero denominator. Final projected vectors undergo masked mean, prefix truncation and only then L2 normalization. Unmasked framing, image and prompt tokens participate in pooling.

3.4 Memory materialization

Q24 boundaries include patch outputs, vision/text Q/K/V, layer outputs, cached PLE, projected soft tokens, fused X0 and normalized embeddings. Scratch regions are prefix-summed in order: V0,V1; vision Q,K, V; soft vectors; fused IDs,positions,masks; X0,X1; PLE; text Q,K,V. Shapes and row-major weight offsets follow E001; unrounded local copies cannot bypass these boundaries.

3.5 Sequencing and faults

Descriptor acceptance enters VALIDATE without a memory request. Valid jobs proceed through separate stage wake, START and WAIT phases. Commands, requests, responses and internal results retire on distinct registered boundaries. Final embedding acceptance precedes a separate completion phase; after completion handshake, the earliest next descriptor is the following edge. Descriptor error 1 precedes extent error 5; subsequently the first detected error wins. Accepted transactions retire before abort completion, and errors produce no embeddings.

4 Verification methods and measured results

4.1 Direct official-weight provenance and independent precision paths

Method. Standard-library parsing validated complete-file and per-tensor SHA-256, BF16 headers, shapes, bounds, interval layout, manifest ranges and finite values; supplied JSON coordinates were cross-checked rather than used as forward weights.

Result. Decoded 55 tensors containing 2,566,786 finite BF16 elements. Software used decoded values with struct-based binary32 operator boundaries and binary64 reductions/math; hardware used nearest-even, signed32-saturating Q8.24 images.

Evidence: E034, E040, E066

4.2 Complete official matrix and exact RTL agreement

Method. Icarus Verilog 12.0 compiled actual RTL with -g2012 -Wall; vvp drove public pins. Seed 17 covered A: three modes at dimensions 6/4/2/1; C: three modes at dimension 12 with 16 vision/24 text iterations; D: three modes at 768/512/256/128; B: vision and joint at dimension 12.

Result. All 29 cases passed with zero output-word mismatches and exact scratch addresses and per-address write histories. B exercised 20 distinct patches on a 4x5 grid, nonzero positions on both axes, multiple vision keys and pooling participant counts 9/6/3/2.

Evidence: E040, E066

4.3 Floating-point similarity and error limits

Method. Compared RTL words divided by 2^24 against independently evaluated software vectors. Required finite nonzero outputs, cosine>=0.999, maximum component error<=0.002, absolute unit-norm error<=0.002 and software/RTL norm difference<=0.002.

Result. All cases passed. B vision was the minimum-cosine and maximum-component-error case: cosine 0.9999999950454138, maximum error 4.7340989112854004e-5, MAE 2.4860103925069172e-5, RMSE 2.8736116710771375e-5 and vector L2-distance 9.954482831057014e-5. Its RTL/software norms were 1.0000000229083967/1.0000000086127838. B joint maximum error was 3.948807716369629e-6. Maximum RTL unit-norm error across the matrix was 4.218036320224883e-8.

Evidence: E066

4.4 Measured per-mode simulated latency

Method. Measured descriptor acceptance to first embedding, final embedding and completion at a 10 ns clock, with always-ready requests/outputs and responses one cycle after request acceptance.

Result. C text first/final/completion: 122580/122591/122594 cycles; vision: 165122/165133/165136; joint: 327389/327400/327403. B vision: 225083/225094/225097; joint: 284579/284590/284593. Completion latencies are respectively 1.22594, 1.65136, 3.27403, 2.25097 and 2.84593 ms. The complete 29-row ledger retains all cycle and nanosecond measurements.

Evidence: E066

4.5 RMS numerical-contract repair without legacy regression

Method. Parameter-aware expectations checked exact epsilon conversion, legacy default, wrapper propagation and L2 independence, followed by the complete functional regression.

Result. The previous multi-patch failure had exact integer agreement but cosine 0.9913286747181946 and maximum error 0.06919039785861969: the legacy epsilon differed from official 1e-6. Selecting the admitted Q80 epsilon resolved the measured discrepancy without changing software epsilon or fixtures. Thirty L2 vectors were identical across epsilon selections; the legacy default and synthetic regression passed.

Evidence: E001, E002, E013, E040, E066

4.6 Retained functional and protocol behavior

Method. Executed the complete synthetic matrix and focused arithmetic, parameter, metadata, reset, backpressure, memory-error, scratch-history, PATCH_SIZE=16 and 280-soft-token fusion checks.

Result. Functional check passed, including legacy numerical behavior and independent protocol-stress coverage. Official benchmark readiness was not substituted for stalled-transaction testing.

Evidence: E030, E038, E040, E066

4.7 Sensitivity to material datapath defects

Method. Compiled and simulated five disposable-source mutations affecting vision positions, soft-token fusion, PLE, full/local scheduling and normalization-before-truncation.

Result. All five mutations were rejected by scratch-history or exact-output assertions; compilation failure or timeout was not counted as detection.

Evidence: E040, E067

4.8 Connected synthesizable structure

Method. Yosys 0.33 processed A and scaled C using hierarchy checking, proc, opt, memory_collect, check -assert, stat and JSON export; inspected reachable modules and epsilon propagation.

Result. Both configurations passed with 33 reachable modules each, 516/555 generic sequential cells and zero latches. Required stages remained connected, with no unresolved black boxes or empty-top substitution. C retained 16/24 layer iterations and the official epsilon override.

Evidence: E040, E068

5 Integration and use

Use rtl/*.sv with top module anvil_e2_top. Read design/ARCHITECTURE.md for the complete tensor-offset, scratch-extent, arithmetic and cycle ledgers; design/OFFICIAL_WEIGHTS_EVALUATION.md explains bounded checkpoint remapping. Its implementation-time pending statements are superseded for measured outcomes by the current functional evidence E066.

Select geometry and RMS_EPSILON_Q80 at elaboration. Provision distinct aligned language-weight, vision-weight, input and scratch regions. Official mappings retain source strides: patch columns channel* 256+dy*16+dx, vision layer 0 reused across iterations, and language layer 0 or 5 selected by the destination schedule. Quantize directly decoded BF16 values with signed32-saturating nearest-even round(value*2^24).

Assert reset, discard canceled transactions, then submit a descriptor only through job_valid/job_ready. Keep stalled memory and output handshakes stable, acknowledge writes as well as reads, consume all embedding beats and accept completion before submitting another job.

Reproduction entry points are python3 tests/verify.py --check functional, python3 tests/verify.py --check sensitivity and python3 tests/verify.py --check synthesis. No arguments execute all three; --check behavior aliases the complete functional check. Dependencies are Python 3 standard library, iverilog, vvp and yosys; NumPy is not required.

The current comparison is evidence/verification/functional/1792082720920298364-239674/official_weights_comparison.json, identified by its adjacent receipt.json. It contains source/input hashes, mappings, parameter overrides, vectors, metrics, geometry and transcript links. Historical fixed-path comparison files are not the current result. Workspace execution was completed; clean-directory portable reproduction was not demonstrated.

6 Limitations and acceptance boundary

Official results use authorized bounded remapping, not native full-checkpoint inference. BOI/EOI public IDs 255999/258882 use available embedding rows 0/1, ordinary tokens use rows 7/19/31, scaled heads pack available coordinates, and PLE rows use the documented scaled selection.

Loaded position shape [2,256,768] derives from a contiguous prefix of original [2,10240,768]. Loaded [1,p,c] therefore maps to original [0,256+p,c], not the native column-axis coordinate [1,p,c].

Matching supplied hashes establishes consistency with mounted provenance, not independent remote checkpoint authentication. BF16 describes weight storage; software is not native BF16 arithmetic or an executable HuggingFace reference implementation.

Parameter defaults describe supported architecture, not a measured full-size workload. Generic synthesis covered A and scaled C only. Simulated latency is not timing closure, default-model throughput, silicon area, power, FPGA deployment or physical sign-off.

Finite regression and five mutation detections do not constitute exhaustive formal proof. No downstream retrieval-quality, semantic-accuracy or application-level benchmark was performed.

A Source inventory

Production inventory: 20 Verilog/SystemVerilog files; 3555 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_e2_address.sv	173	164
rtl/anvil_e2_attention.sv	347	344

Production file	Physical lines	Code lines
rtl/anvil_e2_clock_gate.sv	13	12
rtl/anvil_e2_divide.sv	136	133
rtl/anvil_e2_hybrid_layer.sv	965	947
rtl/anvil_e2_mac.sv	35	35
rtl/anvil_e2_matvec.sv	102	102
rtl/anvil_e2_memory.sv	65	65
rtl/anvil_e2_multimodal_fusion.sv	149	149
rtl/anvil_e2_nonlinear.sv	286	279
rtl/anvil_e2_norm.sv	229	224
rtl/anvil_e2_pool.sv	175	174
rtl/anvil_e2_pool_norm.sv	41	41
rtl/anvil_e2_rope.sv	178	178
rtl/anvil_e2_series.sv	73	70
rtl/anvil_e2_text_frontend_ple.sv	41	41
rtl/anvil_e2_top.sv	298	298
rtl/anvil_e2_vision_encoder_layer.sv	47	47
rtl/anvil_e2_vision_patch_embed.sv	124	124
rtl/anvil_e2_vision_pool_proj.sv	128	128

B Verification record index

Recorded check	Outcome	Execution time (s)
functional	passed	412.975
sensitivity	passed	8.900
synthesis	passed	66.373

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/OFFICIAL_WEIGHTS_EVALUATION.md
E013	rtl/anvil_e2_norm.sv
E030	tests/model.py
E034	tests/official_weights.py
E038	tests/stages.py
E040	tests/verify.py
E066	.vulcan/engineering/run-0002/attempt-0003/stdout.log
E067	.vulcan/engineering/run-0002/attempt-0004/stdout.log
E068	.vulcan/engineering/run-0002/attempt-0005/stdout.log

C Document provenance

Engineering was independently accepted for the recorded source and evidence identity. This is not exhaustive proof, hardware deployment or physical signoff. The companion REPORT.md records the exact report

approval and source manifest, and links the English/Chinese journeys containing original failures and interventions. A report number identifies a generation attempt, not a new hardware revision.

No developer attribution was supplied; none has been inferred.